

FULL STACK TECHNOLOGIES

LAB 4

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TOPICS COVERED:

- Node.js concepts - Server, file handling ,url, querystring, loading data from json and writing data into json, downloading file.

ADD.HTML

```
<> add.html X
labc > <> add.html > style
1 <div><form method="post">
2 Enter Title <br><input type="text" name="title"><br>
3 Enter Author<br> <input type="text" name="author"><br><br>
4 <button type="button" onclick="addNewAuthor()">Add More Authors</button>
5 <input type="submit" value="Add Book">
6 </form>
7 </div>
8 <script>
9     function addNewAuthor(){
10         const a=document.querySelector('button');
11         a.insertAdjacentHTML('beforebegin','<input type="text" name="author"><br><br>');
12     }
13 </script>
14 <style>
15     div{
16         text-align:center;
17         padding-top:50px;
18     }
19 </style>
```

BOOK.HTML

```
<> add.html <> book.html X
labc > <> book.html > table
1 <table border="1"><tr><th>Title</th><th>Author</th><th>Download Link</th><tr><td>HP</td><td>AA,BB,CC</td><td><a href="
2 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/1.txt" download="HPP.txt">Download</a></td></tr>
3 <tr><td>nbl</td><td>jfl,nbl,mknl</td><td><a href="./download/2.txt" download="nbl.txt">Download</a></td></tr>
4 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/3.txt" download="HP.txt">Download</a></td></tr>
5 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/4.txt" download="HPP.txt">Download</a></td></tr>
6 <tr><td>mjbv</td><td>jbvk.fd,m,kbnc</td><td><a href="./download/5.txt" download="mjbv.txt">Download</a></td></tr>
7 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/6.txt" download="HP.txt">Download</a></td></tr>
8 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/7.txt" download="HPP.txt">Download</a></td></tr>
9 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/8.txt" download="HP.txt">Download</a></td></tr>
10 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/9.txt" download="HPP.txt">Download</a></td></tr>
11 <tr><td>nxbm</td><td>mcx,,m, zbx,,kj cdvk</td><td><a href="./download/10.txt" download="nxbm.txt">Download</a></td></tr>
12 </table>
```

LIBRARY.JS

```
labc > labexp > JS library.js > getbookauthor
1  const a=require('fs')
2  let books=[]
3  let data=a.readFileSync('./datas.json')
4  books=JSON.parse(data.toString())
5  var i=books.length
6  function addbook(title,author){
7      books[i]={title:title,author:author,filename:`${i}.txt`}
8      a.writeFile(`./books/${i}.txt`,`${title},${author}`.repeat(1000000),()=>{ })
9      i++
10     a.writeFileSync("datas.json",JSON.stringify(books),()=>>{ })
11 }
12 function getbookauthor(title)
13 {
14     for(let j=0;j<i;j++)
15     {
16         if(books[j].title==title)
17         {
18             return books[j].author;
19         }
20     }
21 }
22 function getbook(title)
23 {
24     for(let j=0;j<i;j++)
25     {
26         if(books[j].title==title)
27         {
28             return books[j].filename;
29         }
30     }
31 }
32 function getbooktitle(author)
33 {
34     var authors=[]
35     for(let j=0;j<i;j++)
36     {
```

```
function getbooktitle(author)
{
    var authors=[]
    for(let j=0;j<i;j++)
    {
        for(let k=0;k<books[j].author.length;k++)
        {
            if(books[j].author[k]==author)
            {
                authors.push(books[j].title)
            }
        }
    }
    return authors
}
function getbookna(){
    var na=[]
    for(let j=0;j<i;j++)
    {
        na[j]={title:books[j].title,author:books[j].author}
    }
    return na;
}
function getTitle(filename)
{
    for(let j=0;j<i;j++)
    {
        if(books[j].filename==filename)
        {
            return books[j].title;
        }
    }
}

module.exports={addbook,getTitle}
addbook("hdgfh",["yk","ll"])
```

SERVER.JS

```
1  const a=require('./library.js')
2  const b=require('http')
3  const c=require('fs')
4  const q=require('querystring')
5  var inf=c.readFileSync("downloads.json")
6  var infdata=JSON.parse(inf.toString())
7  var downloads=[]
8  const server=b.createServer((req,res)=>{
9      var data=c.readFileSync("./datas.json")
10     let books=JSON.parse(data.toString())
11     c.writeFileSync("./book.html",`<table border='1'><tr><th>title</th><th>author</th><th>download link</th></tr>`)
12     for(let j=0;j<books.length;j++)
13     {
14         c.appendFileSync("./book.html",`<tr><td>${books[j].title}</td><td>${books[j].author}</td><td><a href='./download/${books[j].filename}'>Download</a></td></tr>`)
15     }
16     c.appendFileSync("./book.html",`</table>`)
17
18     if(req.url=="/add" && req.method=="POST")
19     {
20         let body=""
21         req.on('data',(chunk)=>{body+=chunk.toString()})
22         req.on('end',()=>{
23             var booksdata=q.parse(body)
24             let author=Array.isArray(booksdata.author)?booksdata.author:[booksdata.author];
25             let title=booksdata.title;
26             a.addbook(title,author)
27             resp.setHeader("Content-Type","text/html")
28             resp.write("<h3>Book added Successfully</h3><a href='./book'>View Books</a>")
29             resp.end()
30         })
31     }
32     return;
33 }
34 let filename
35 switch(req.url)
36 {
```

```

35 switch(req.url)
36 {
37     case '/book':
38     {
39         filename="./book.html"
40         resp.statusCode=200;
41         break;
42     }
43     case '/add':
44     {
45         filename="./add.html"
46         resp.statusCode=200;
47         break;
48     }
49     /*case '/download':
50     {
51         filename="./download.html"
52         resp.statusCode=200;
53         break;
54     }*/
55     default:{
56         filename="./err.html"
57         resp.statusCode=404;
58         break;
59     }
60 }
61 if(req.url.startsWith('/download/'))

```

```

55     default:{
56         filename="./err.html"
57         resp.statusCode=404;
58         break;
59     }
60 }
61 if(req.url.startsWith('/download/'))
62 {
63     filename="./books/"+req.url.substring(10);
64     resp.setHeader("Content-Type","text/plain")
65     resp.setHeader("content-disposition","attachment;filename=${a.getTitle(req.url.substring(10))}.txt");
66     resp.statusCode=200;
67     const rs=c.createReadStream(filename);
68     rs.pipe(resp);
69     downloads++;
70     infdata.total++;
71     c.writeFileSync("downloads.json",JSON.stringify(infdata))
72     return;
73 }
74 resp.setHeader("content-Type","text/html")
75 c.readFile(filename,(err,data)=>{if(err){resp.write("error")})
76 else{resp.write(data)}
77 })
78 resp.end()
79 })
80 })

```

DATA.JSON

```

add.html {} data.json X
bc > {} data.json > ...
1 [{"title":"HP","author":["AA","BB","CC"],"filename":"0.txt"},{"title":"HPP","author":["DD","BB","CC"],"filename":"1.txt"},{"title":"nblk","author":["

```

DOWNLOADINFO.JSON

```

<> add.html {} downloadinfo.json X
labc > {} downloadinfo.json > ...
1 [{"totalDownloads":2}]

```

OUTPUT:

localhost:345/book

Title	author	download link
hdgf	yk,ll	Download
mfdn	c m,mdkvbj	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download

localhost:345/add

localhost:345/add

Enter Title

vcx,k

Enter Author

vdsjkh

vbdnjkwhs

vdbsjh

Add More Authors

Add Book

On clicking download ,file gets downloaded

localhost:345/book

Title	Author	Download Link
HP	AA.BB.CC	Download
HPP	DD.BB.CC	Download
nblk	jfl,nbkl,mknbl	Download
HP	AA.BB.CC	Download
HPP	DD.BB.CC	Download
mjkbv	jbv.k.fid,m.klbn	Download
HP	AA.BB.CC	Download
HPP	DD.BB.CC	Download
HP	AA.BB.CC	Download
HPP	DD.BB.CC	Download
mxcbum	mxc,m,zbx,kj edvk	Download
vcx,k	vdsjkh,vbdnjkwhs,vdbsjh	Download

mjkbv.txt
1.953 KB • Done

